



Aditya Vibhute

Energy Science and Engineering

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Dual Degree (B.Tech. + M.Tech.)

LinkedIn

Examination	Institute	Year	CPI
Graduation	IIT Bombay	2026	8.01

MASTER'S THESIS

Sizing of BESS for Industrial Peak Shaving | *Master's Thesis* | *DESE* (Aug '25 - Jun '26)

Guide: Prof. D. Venkatramanan

- Built a **linear-programming optimization model** in **Python** (HiGHS solver) for battery dispatch and demand-charge peak shaving across **10 real industrial load profiles**
- Engineered a **data pipeline** (15-min for dispatch, 1-min for sizing) to process large time-series datasets
- Ran a full **techno-economic analysis** (NPV, IRR, discounted payback) under real tariff structures, quantifying project viability across profiles and the sensitivity to demand-charge penalties
- Modelled **LFP battery degradation** and integrated it into the lifetime and replacement-cost analysis

PROFESSIONAL EXPERIENCE

BESS Techno-Commercial Analysis | *Research Intern* | *IIT Bombay* (May '24 - Jul '24)

Guide: Prof. P. Sunthar

- Derived closed-form **savings expressions** for different BESS strategies and system configurations
- Developed **LCoE models** for BESS, standalone or with PV, to cut grid and DG-set dependence
- Built a **web framework** packaging these models into an application for industry use

KEY PROJECTS

Design of Photovoltaic Power Plants | *Course Project* | *EE770* (Jan '25 - Apr '25)

Guide: Prof. Narendra Shiradkar, Department of Electrical Engineering

- Simulated fixed, single-axis, and dual-axis solar tracking systems across different locations using **NREL's SAM**
- Conducted **LCOE-based techno-economic analysis** to determine the financial viability of tracking mechanisms
- Established that **dual-axis** achieved **67% higher energy yield** and **27.7% lower LCOE** than fixed systems

Simulation of Solar Mini-Grid System | *Course Project* | *TD619* (Jan '25 - Apr '25)

Guide: Prof. Anand Rao, CTARA

- Designed **3.52kW PV-battery system** based on real-world load demand from past **3 yrs** for rural electrification
- Mapped energy consumption from appliances used in **15+** households for sizing a PV-battery system in **HOMER**
- Performed optimization across multiple PV-battery configurations to identify cost savings of **10%** in LCOE

Grid-connected Rooftop Solar PV analysis | *Course Project* | *TD619* (Jan '25 - Apr '25)

Guide: Prof. Anand Rao, CTARA

- Simulated a **2.5 kW** rooftop PV system in **HOMER** with a **4.83-year** discounted payback
- Analyzed and evaluated the enablers and barriers to scaling grid-connected rooftop-PV adoption across India

Modelling of sCO₂ Power Cycle | *Course Project* | *EN317* (Aug '23 - Nov '23)

Guide: Prof. Anish Modi, DESE

- Modelled a **150 MW sCO₂ Brayton cycle** using **Python + CoolProp**, achieving **39.16% thermal efficiency**
- Designed & sized microchannel LTR with **1.85 million channels**, and **21,360 sq.m** exchange area in Python

Solar Thermal System for Industrial Process Heat | *Course Project* | *EN630* (Jan '25 - Apr '25)

Guide: Prof. Anish Modi, DESE

- Designed a **129 sq.m** Parabolic Dish Solar Collector system achieving a thermal output of **57 kW** using Python
- Engineered a solar thermal solution that operates with an efficiency of **76.19%**, with temperatures up to **270°C**
- Calculated an annual energy savings of **522.88 GJ**, abating **31,260 kg** of carbon dioxide emissions per year

Inductor Design for boost PFC circuit | *Course Project* | *EN665* (Aug '24 - Nov '24)

Guide: Prof. Ravi Prakash Reddy, DESE

- Designed and implemented a **DC-DC PFC boost converter** achieving **95.4%** efficiency and **0.713** power factor
- Fabricated a custom **881 μ H inductor** using EE 42/21/20 core based on **area product** and **MMF calculations**
- Verified DCM mode operation with **< 3.5%** voltage ripple, simulating 82V output converter operation in PLECS

Design of Sun Tracking System | Course Project | EN408

(Jan '25 - Apr '25)

Guide: Prof. Rajesh Gupta, DESE

- Performed torque calculations accounting for **moment of inertia** and **wind loads** to ensure robust motor selection
- Designed an **open-loop motor control system** using a high-torque Stepper Motor for precise solar tracking
- Developed an analytical **astronomical tracking algorithm** to control motor actuation from sunrise to sunset

Design of Synchronous Reluctance Machine | Seminar | EN406

(Aug '24 - Nov '24)

Guide: Prof. V.S.S. Pavan Kumar Hari, DESE

- Reviewed modern design methodologies for Synchronous Reluctance Machines and analysed multiple rotor and stator configurations for their torque, ripple, and power-factor trade-offs
- Benchmarked the SynRM against an induction motor on efficiency, losses, torque, and cost

Self-Powered Gas Pipeline Leak Detection System | Course Project | EN405 (Aug '24 - Nov '24)

Guide: Prof. Sandeep Kumar, DESE

- Engineered a **4.5W** wind energy harvesting setup using a **5cm HAWT**, delivering up to **28V at 20m/s flow**
- Integrated **pressure sensor** and **4G LTE module**, for autonomous leak detection and wireless data transmission
- Achieved accurate leak alerts on **pressure drops** $< 1\text{hPa}$ while performing open/closed pipe experiments

TECHNICAL SKILLS

Modelling & Optimization	Python (NumPy, Pandas, SciPy), Linear Programming & optimization (HiGHS), techno-economic (NPV/IRR/LCoE) & dispatch modelling, statistical & numerical analysis
Tools	MATLAB, HOMER Pro, NREL SAM, CoolProp, PLECS, OpenFOAM, Git, L ^A T _E X; C/C++

KEY COURSES

- Maths courses:** Linear Algebra, Differential Equations, Calculus-I & II, Numerical Analysis
- Electrical courses:** Electrical Machines, Power Electronics, Control & Instrumentation, Electrical Energy Systems, Energy Systems Modelling & Analysis, EV Powertrains, Power Converters for EV Charging, Electric Distribution Systems, Design & Evaluation of Photovoltaic Power Plants, Power Electronic Converters for Renewable Energy, Application of Power Electronics to Power Systems, Power Generation & System Planning, Modeling & Control of Inverter-based Renewable Resources
- Energy courses:** Energy Policy & Planning, Energy Resources Economics & Environment, Energy Management, Thermodynamics & Energy Conversion, Utilisation of Solar Thermal Energy, Wind Energy Conversion Systems, Conservation of Energy in Buildings
- Lab courses:** Basic Electrical Engineering Lab, Electrical Machines & Power Electronics Lab, Thermal & Fluid Engineering Lab, Solar Energy Lab, IC Engine & Combustion Lab

POSITIONS OF RESPONSIBILITY

Teaching Assistant | Dept of Energy Science and Engineering | EN327

(Aug '25 - Nov '25)

Prof Pavan Kumar Hari & Prof D Venkatramanan

- Appointed as Teaching Assistant for **Power electronics and Instrumentation lab** for third year energy students
- Helping in the understanding of working of **linear potentiometer** and **linear voltage potential transducer**

Teaching Assistant | Dept of Energy Science and Engineering | EN224

(Jan '26 - Apr '26)

Prof Pavan Kumar Hari & Prof D Venkatramanan

- Appointed as Teaching Assistant for **Electrical Networks and Machines Lab** for second year energy students
- Mentoring **50+** students for their lab experiments and evaluate their work along with testing their understanding

SCHOLASTIC ACHIEVEMENTS

- Awarded the highest grade (**AA**) in both Dual Degree Project stages, Energy Innovation Lab, Seminar, and the Thermal & Fluid Engineering Lab, with **10+** ABs
- Achieved semester **SPIs up to 10.0**, with **AA** grades in both stages of the Dual Degree Project
- Cleared the preliminary round of the **National Mathematics Olympiad** (AISMTA)

EXTRACURRICULAR ACTIVITIES

Tech	Made 4+ single-player games and 2+ web apps for academic tasks; completed the AI project ComicGPT (Seasons of Code); boot-camps on IoT and house design
Sports	Awarded Best Cadet at an NCC training camp; competed in hostel-level 8-ball pool billiards
Cultural	Won 10+ awards for handwriting, essay, and story writing; cleared Grade 1 music theory (NSO Guitar); scriptwriter and performer for the school dramatics club
Misc.	Volunteered in a social-service initiative for special children ('Sunrise'); cleared 5+ IMO / NSO Olympiads